

Domains and Ranges

EXAMPLE 1 (a) These are functions of two variables. Note the restrictions that may apply to their domains in order to obtain a real value for the dependent variable z .

Function	Domain	Range
$z = \sqrt{y - x^2}$	$y \geq x^2$	$[0, \infty)$
$z = \frac{1}{xy}$	$xy \neq 0$	$(-\infty, 0) \cup (0, \infty)$
$z = \sin xy$	Entire plane	$[-1, 1]$

(b) These are functions of three variables with restrictions on some of their domains.

Function	Domain	Range
$w = \sqrt{x^2 + y^2 + z^2}$	Entire space	$[0, \infty)$
$w = \frac{1}{x^2 + y^2 + z^2}$	$(x, y, z) \neq (0, 0, 0)$	$(0, \infty)$
$w = xy \ln z$	Half-space $z > 0$	$(-\infty, \infty)$

DEFINITION We say that a function $f(x, y)$ approaches the **limit** L as (x, y) approaches (x_0, y_0) , and write

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f ,

$$|f(x, y) - L| < \epsilon \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

THEOREM 1—Properties of Limits of Functions of Two Variables

The fol-

lowing rules hold if L , M , and k are real numbers and

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

1. *Sum Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) + g(x,y)) = L + M$$

2. *Difference Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) - g(x,y)) = L - M$$

3. *Constant Multiple Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} kf(x,y) = kL \quad (\text{any number } k)$$

4. *Product Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \cdot g(x,y)) = L \cdot M$$

5. *Quotient Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}, \quad M \neq 0$$

6. *Power Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^n = L^n, \quad n \text{ a positive integer}$$

7. *Root Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n},$$

n a positive integer, and if n is even, we assume that $L > 0$.

EXAMPLE 1 In this example, we can combine the three simple results following the limit definition with the results in Theorem 1 to calculate the limits. We simply substitute the x and y values of the point being approached into the functional expression to find the limiting value.

$$(a) \quad \lim_{(x,y) \rightarrow (0,1)} \frac{x - xy + 3}{x^2y + 5xy - y^3} = \frac{0 - (0)(1) + 3}{(0)^2(1) + 5(0)(1) - (1)^3} = -3$$

$$(b) \quad \lim_{(x,y) \rightarrow (3,-4)} \sqrt{x^2 + y^2} = \sqrt{(3)^2 + (-4)^2} = \sqrt{25} = 5$$

DEFINITIONA function $f(x, y)$ is **continuous at the point** (x_0, y_0) if1. f is defined at (x_0, y_0) ,2. $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y)$ exists,3. $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = f(x_0, y_0)$.A function is **continuous** if it is continuous at every point of its domain.

Limits with Two Variables

Find the limits in Exercises 1–12.

1. $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 - y^2 + 5}{x^2 + y^2 + 2}$

2. $\lim_{(x,y) \rightarrow (0,4)} \frac{x}{\sqrt{y}}$

3. $\lim_{(x,y) \rightarrow (3,4)} \sqrt{x^2 + y^2 - 1}$

4. $\lim_{(x,y) \rightarrow (2,-3)} \left(\frac{1}{x} + \frac{1}{y} \right)^2$

5. $\lim_{(x,y) \rightarrow (0,\pi/4)} \sec x \tan y$

6. $\lim_{(x,y) \rightarrow (0,0)} \cos \frac{x^2 + y^3}{x + y + 1}$

Limits of Quotients

Find the limits in Exercises 13–24 by rewriting the fractions first.

13. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x^2 - 2xy + y^2}{x - y}$

14. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x^2 - y^2}{x - y}$

15. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq 1}} \frac{xy - y - 2x + 2}{x - 1}$

16. $\lim_{\substack{(x,y) \rightarrow (2,-4) \\ x \neq -4, x \neq x^2}} \frac{y + 4}{x^2y - xy + 4x^2 - 4x}$

17. $\lim_{\substack{(x,y) \rightarrow (0,0) \\ x \neq y}} \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$

18. $\lim_{\substack{(x,y) \rightarrow (2,2) \\ x+y \neq 4}} \frac{x + y - 4}{\sqrt{x + y} - 2}$

19. $\lim_{\substack{(x,y) \rightarrow (2,0) \\ 2x-y \neq 4}} \frac{\sqrt{2x - y} - 2}{2x - y - 4}$

20. $\lim_{\substack{(x,y) \rightarrow (4,3) \\ x \neq y+1}} \frac{\sqrt{x} - \sqrt{y+1}}{x - y - 1}$

21. $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2}$

22. $\lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(xy)}{xy}$

23. $\lim_{(x,y) \rightarrow (1,-1)} \frac{x^3 + y^3}{x + y}$

24. $\lim_{(x,y) \rightarrow (2,2)} \frac{x - y}{x^4 - y^4}$

Limits with Three Variables

Find the limits in Exercises 25–30.

25. $\lim_{P \rightarrow (1,3,4)} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$

26. $\lim_{P \rightarrow (1,-1,-1)} \frac{2xy + yz}{x^2 + z^2}$

27. $\lim_{P \rightarrow (\pi,\pi,0)} (\sin^2 x + \cos^2 y + \sec^2 z)$

28. $\lim_{P \rightarrow (-1/4,\pi/2,2)} \tan^{-1} xyz$

29. $\lim_{P \rightarrow (\pi,0,3)} ze^{-2y} \cos 2x$

30. $\lim_{P \rightarrow (2,-3,6)} \ln \sqrt{x^2 + y^2 + z^2}$